

The HEIC Conversion Tax: Quantifying the File-Size Cost of Converting iPhone HEIC Photos to JPEG and PNG at Matched Perceptual Quality

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Abstract

Converting Apple HEIC photographs to JPEG is one of the most common image operations performed by consumers: the query "heic to jpg" attracts on the order of 246,000 searches per month. Yet the file-size consequence of this conversion is rarely quantified. We benchmark a corpus of photographs encoded to HEIC at several iOS-like quality levels and then converted, as typical online tools do, to JPEG (across a quality ladder) and to lossless PNG. For each output we measure the structural similarity index (SSIM) against the lossless master, allowing a comparison of file size *at matched perceptual quality*. Converting HEIC to JPEG inflates the corpus by a median factor of **1.75× to 4.65×**, and to PNG by **3.87× to 13.60×**, with no gain in perceptual quality. We conclude that the conversion is a pure storage cost imposed by ecosystem compatibility rather than any quality benefit, and we release the full dataset and reproducible harness.

1. Introduction

Since iOS 11 (2017), Apple devices capture photographs by default in the High Efficiency Image File Format (HEIF) with HEVC-coded image items, commonly seen by users through the `.heic` extension [2]. HEIF/HEVC intra-coding is markedly more efficient than baseline JPEG. However, HEIC is not universally supported by web platforms, editing software, and third-party services, so users routinely convert their photos to JPEG (or, less advisedly, PNG) for compatibility.

The performance of HEVC intra-coding relative to JPEG is well studied in the coding literature, but consumers experience the conversion as a black box: they know they *must* convert, but not what it costs. This paper measures that cost directly on a photographic corpus and expresses it at matched perceptual quality so that size differences cannot be attributed to a quality change.

2. Method

Corpus. We use a set of lossless photographic masters (PNG). For each master we produce a HEIC file at iOS-like quality settings using `heif-enc` (libheif, x265 backend) at four quality points, approximating

"the file already on the phone."

Conversions. From each HEIC we generate the outputs a typical converter produces: JPEG at quality {80, 90, 95, 100} using the same imaging engine that backs the authors' browser tools (libvips/sharp), and lossless PNG.

Quality metric. For the HEIC and every JPEG output we compute the structural similarity index, SSIM [1], against the lossless master using `ffmpeg`. PNG is lossless by construction. Reporting size at matched SSIM ensures that a smaller file is not merely a lower-quality file.

Measure of the "tax." For each master and HEIC quality level we compute the ratio of the converted file size to the source HEIC size, and report the median ratio across the corpus (the "size tax"), alongside the mean SSIM of the HEIC relative to the master.

The complete harness (`benchmarks/heic-tax-benchmark.mjs`) and the resulting per-image and summary CSVs are released with this preprint (Section 6).

3. Results

Table 1 reports, for each iOS-like HEIC quality point, the mean SSIM of the HEIC relative to the lossless master and the median size multiplier incurred by converting to JPEG (at matched perceptual quality) and to PNG.

Table 1. HEIC-to-JPEG and HEIC-to-PNG size tax at matched perceptual quality.

HEIC quality	Mean SSIM (HEIC vs master)	JPEG size tax (median)	PNG size tax (median)
Lower (q40)	0.931	4.65×	13.60×
(q50)	0.961	3.47×	8.46×
(q60)	0.978	2.51×	5.50×
Higher (q70)	0.987	1.75×	3.87×

Two patterns are consistent across the corpus. First, at every quality level the JPEG that matches the HEIC's perceptual quality is substantially larger than the HEIC — from 1.75× at high quality to 4.65× at lower quality — confirming that the conversion buys no quality and only costs space. Second, converting to PNG, a choice many users make believing it to be "safer" or "higher quality," inflates photographic content enormously (up to 13.6×) because lossless entropy coding is ill-suited to natural-image data.

4. Discussion

The results reframe a mundane task as a measurable, ecosystem-imposed storage cost. For an individual photo the difference is invisible; aggregated across a photo library, a device, or a content pipeline, the multiplier is the difference between, for example, a 2 GB and a 6–9 GB export.

Two practical recommendations follow. (1) When JPEG is required for compatibility, quality ~90 is sufficient to match HEIC perceptually for most images; higher settings mostly add bytes. (2) PNG should not be used as a "convert-for-safety" target for photographs; it is appropriate for graphics and screenshots, not camera images. Where the destination platform supports it, converting to AVIF or WebP rather than JPEG recovers most of the efficiency lost by leaving HEIC.

Limitations. SSIM is a widely used perceptual proxy but not a substitute for subjective testing; differences above SSIM 0.99 approach the metric's noise floor. Absolute sizes depend on the corpus; the reported multipliers are the generalizable quantity and will vary with image content and encoder settings, which is precisely why the harness is released for re-running on any corpus.

5. Conclusion

Converting iPhone HEIC photographs to JPEG or PNG, an operation performed at massive scale, imposes a median $1.75\times$ – $4.65\times$ (JPEG) or $3.87\times$ – $13.60\times$ (PNG) file-size penalty at matched perceptual quality, with no quality benefit. The "HEIC tax" is a cost of compatibility, and it is avoidable wherever modern formats are accepted downstream.

6. Data and code availability

The full per-image dataset (`heic-tax-benchmark.csv`), the summary (`heic-tax-summary.csv`), and the reproducible harness (`heic-tax-benchmark.mjs`) are openly available under CC BY 4.0 (data) and MIT (code) at <https://github.com/cleanor-app/cleanor-storage-lab>. A versioned, DOI-assigned archive is deposited on Zenodo. A human-readable write-up with interactive charts is at <https://cleanor.app/blog/heic-to-jpg-conversion-file-size-tax-benchmark>.

References

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